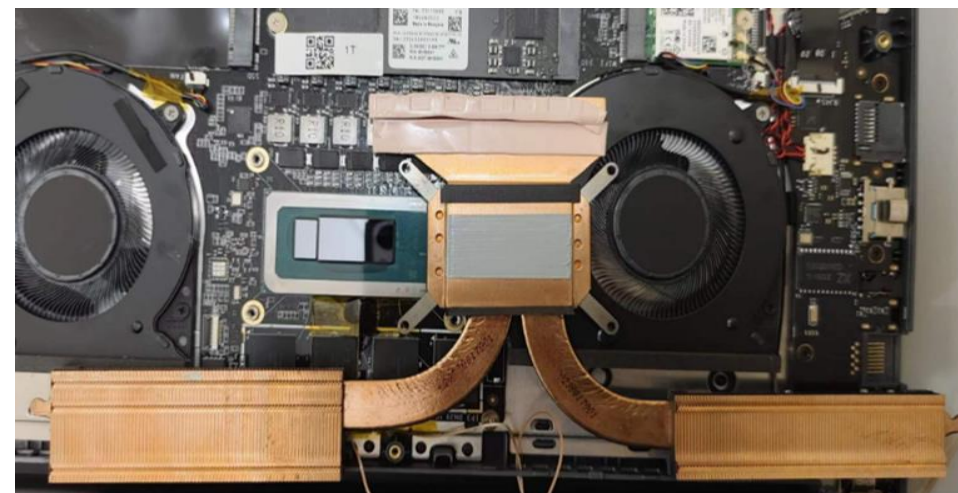
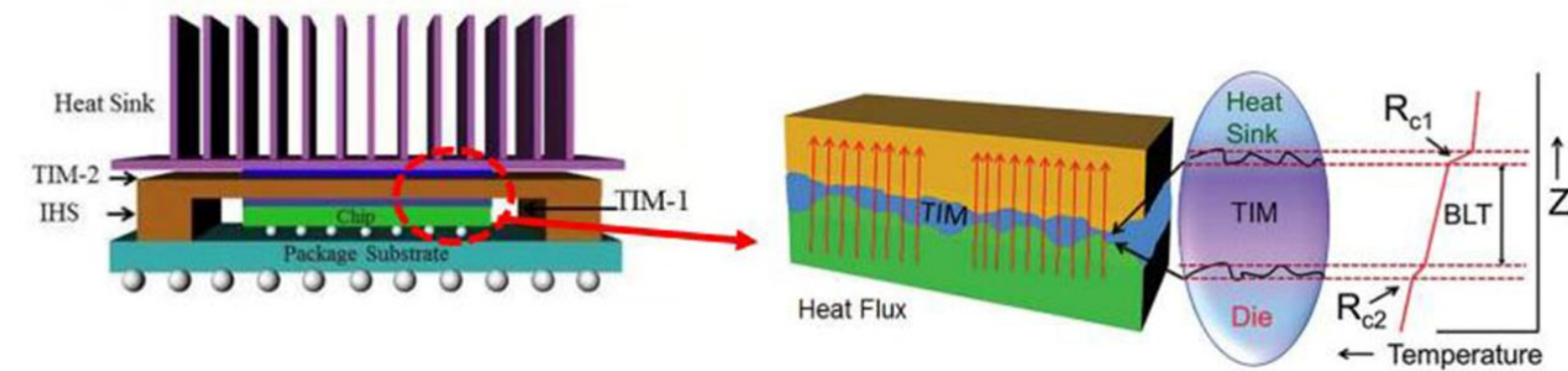


Deep-Materials Insulating Pads



Use fundamental science to make practical improvements in TIM

Deep-Materials

TIM Product Brochure

Use *fundamental science* to make practical improvements in Thermal Materials

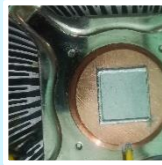
CPU/GPU TIMs



Grease

- Best in class performance
- Significantly reduce pump out

Highest performance traditional grease & Innovative Liquid Metal grease with low resistance



Liquid Metal

- Multi-phase Liquid metal – high viscosity
- Practical to use
- Highest performance

Brand new category of TIM with perf. of LM but ease of grease

Gap Fillers



Gels

- Highest conductivity
- Almost no oil seepage

Many kinds of unique gels – e.g. ultra low oil seepage gel, low dielectric constant gels



Gap Pads

- Almost eliminate oil seepage
- Reliable to very high temperatures

Ultra low oil seepage & ultra low outgassing silicone pads

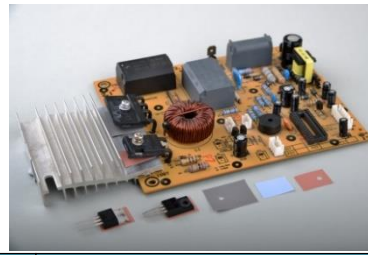
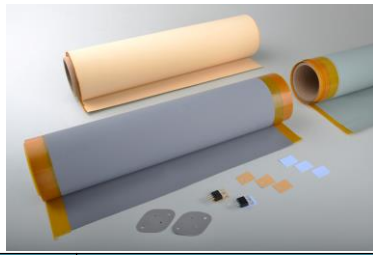
GAP Fillers

- For Applications in lower power/thicker TIM applications
- Do not require high pressure or mechanical support
- Typically used in consumer electronics, SSDs, Radios
- Performance measured in thermal conductivity (W/mK)

Our offerings in this space

- Liquid Gap Fillers/Gels
 - Single part (Fully cured)/Two part (in situ cure)
- Gap pads

Insulating Pads



Applications:
Power supplies, Power electronics, UPS, Inverters,
High voltage industry, Automotive

	TIP-1000	TIP-1600	TIP-3500	PCM 2500-PI	Test Methods
Color	Gray	Pink/Gray	White	Gray	Visual
Hardness (Shore A)	80	80	80	/	ASTM D2240
Thickness (mm)	0.229	0.203	0.25	0.127/0.205/0.254	- -
Elongation at Break (%)	2.5	2.5	2.5	40	ASTM D412
Tensile Strength (psi/MPa)	1300/9	2880/20	2880/20	7000/50	ASTM D412
Continuous Use Temp (°C)	-60 to 180	-60 to 180	-60 to 200	-40 to 130	--
Phase Change Temperature	/	/	/	50 °C	DSC
Reinforcement Carrier	Fiberglass	Fiberglass	Fiberglass	PI film	/
Electrical Properties					
Dielectric Constant (1000Hz)	6	6	6	4.5	ASTM D150
Dielectric Breakdown Voltage (Vac)	5000	5000	5000	5000	ASTM D149
Volume Resistivity (ohm-meter)	10 ¹⁰	10 ¹⁰	10 ¹⁰	10 ¹⁰	ASTM D257
Flammability Rating	V-0	V-0	V-0	V-0	UL94
Thermal Properties					
Thermal Conductivity (W/m·K)	1.0	1.6	3.5	2.5	ASTM D5470
Thermal Impedance (°C-in²/W @ 50 psi)	1.10	0.62	0.33	0.15/0.17/0.18	ASTM D5470

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